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Diversity and prevalence of *Bartonella* species in small mammals from Slovakia, Central Europe 2

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Abstract Wild-living rodents are important hosts for zoonotic pathogens. *Bartonella* infections are widespread in rodents; however, in Slovakia, knowledge on the prevalence of these bacteria in small mammals is limited. We investigated the prevalence and diversity of *Bartonella* species in the spleens of 640 rodents of six species (*Apodemus flavicollis*, *Apodemus sylvaticus*, *Myodes glareolus*, *Microtus arvalis*, *Microtus subterraneus*, and *Micromys minutus*) and in the European mole (*Talpa europaea*) from three different habitat types in south-western and central Slovakia. Overall, the prevalence of *Bartonella* spp. in rodents was 64.8%; a rate of 73.8% was found in natural habitat (deciduous forest), 56.0% in suburban forest park and 64.9% in rural habitat. *Bartonella* spp. were detected in 63.0% of *A. flavicollis*, 69% of *My. glareolus* and 61.1% of *M. arvalis* and in *T. europaea*. However, *Bartonella* were not found in the other examined rodents. Molecular analyses of the 16S–23S rRNA intergenic spacer region revealed the presence of four different *Bartonella* spp. clusters. We identified *B. taylorii*, *B. rochalimae*, *B. elizabethae*, *B. grahamii* and *Bartonella* sp. wbs11 in *A. flavicollis* and *My. glareolus*. *Bartonella* genotypes ascribed to *B. taylorii* and *B. rochalimae* were found in *M. arvalis*. *B. taylorii* was identified in *T. europaea*. Questing *Ixodes ricinus* ticks that were collected at the study sites were not infected with *Bartonella*. This study improves our understanding of the ecoepidemiology of *Bartonella* spp. in Europe and underlines the necessity for further research on *Bartonella*-host-vector associations and their consequences on animal and human health in Slovakia. 6 7

Keywords *Bartonella* · Rodents · Slovakia 8

Introduction 9

Bartonella, previously known as *Grahamella* (Birtles et al. 1995), are Gram-negative bacteria of medical and veterinary importance that belong to the class of alpha-proteobacteria. They infect the erythrocytes and endothelial cells of mammals and are adapted to distinct mammalian hosts. *Bartonella* are transmitted by ectoparasitic arthropods, although knowledge of their vectors is not well understood (Regier et al. 2016). Several blood-feeding arthropods, such as sand flies, human body lice, fleas, vole ear mites and ticks have been implicated in *Bartonella* transmission (Breitschwerdt and Kordick 2000). Some species of *Bartonella* (e.g. *B. henselae*, *B. elizabethae*, *B. alsatica*, *B. vinsonii* subsp. *arupensis*, *B. bacilliformis*, *B. quintana*, *B. grahamii* and *B. washoensis*) have been associated with human diseases with presentations that range from subclinical and self-limiting infection to severe, life-threatening disease and have an increased risk of infection in immunocompromised patients (Breitschwerdt and Kordick 2000; Mogollon-Pasapera et al. 2009; Mosepele et al. 2012; Regier et al. 2016). Clinical studies have supported the transmission of *Bartonella* by ticks to humans (Morozova et al. 2005; Billeter et al. 2008, 2012). 10

Currently, 46 *Bartonella* species are listed in the Taxonomy Database of the National Center for Biotechnology 11

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Information (www.ncbi.nlm.nih.gov). *Bartonella* phylogeny, which is based on a genome-wide dataset, has revealed four lineages. During evolution, these lineages have adapted to a variety of hosts and developed virulence factors associated with a diverse set of symptoms (Mullins et al. 2017). In contrast to the ancestral lineage 1, lineages 2, 3 and 4 comprise species isolated from various hosts. While the species of lineage 2 are limited to ruminant hosts and have an overlapping host range, the diversification of lineages 3 and 4 appear to result from the adaptation to distinct mammalian hosts (including voles, mice, rats, red squirrels, cats, dogs, foxes, rabbits and humans). Taxa of lineage 4 comprise a wide range of divergently adapted species that are frequently found in free-living host populations. In contrast, the taxa of lineage 3, except for *B. clarridgeiae*, were sampled at a low prevalence (Engel et al. 2011). 1

Several *Bartonella* species have been detected in rodents from various regions worldwide (Gutiérrez et al. 2015). Among them, *B. elizabethae*, *B. tribocorum*, *B. grahamii*, *B. vinsonii* subsp. *arupensis* and *B. washoensis* have been associated with human infections (Daly et al. 1993; Welch et al. 1999; Kosoy et al. 2003, 2010). These findings underline the importance of understanding the reservoir competence of rodents for *Bartonella*, which is limited in Slovakia. 2

The yellow-necked mouse, *Apodemus flavicollis* (Melchior, 1834), and the bank vole, *Myodes glareolus* (Schreber, 1780), belong to predominantly occurring wild-living rodents in Slovakia. The striped field mouse, *Apodemus agrarius* (Pallas, 1771), is found primarily in the eastern part of the country. These rodents have a small home range and a relatively short lifespan. *Apodemus* spp., *Microtus* spp. and *My. glareolus* are hosts to ectoparasitic arthropods that play important roles in the maintenance and circulation of tick-borne microbial agents, such as *Borrelia burgdorferi* s.l., *Borrelia miyamotoi*, *Anaplasma/Ehrlichia*, *Rickettsia* spp., *Babesia microti* and Candidatus Neoehrlicha mikurensis (Rizzoli et al. 2014). In Slovakia, the reservoir competence of wild-living rodents has been identified for several tick-borne zoonotic pathogens, i.e. *Borrelia afzelii* (Hanincová et al. 2003), Cand. Neoehrlichia mikurensis (Blaňarová et al. 2016; Hamšíková Svitálková et al. 2016), *Babesia microti* (Blaňarová et al. 2016; Hamšíková et al. 2016), and *Borrelia miyamotoi* (Hamšíková et al. 2017). However, the reservoir role of rodents for rickettsiae (Minichová et al. 2017) and the role of ticks as *Bartonella* vectors have not been identified. 3

Šebek (1960, 1975) was the first to report *Bartonella* infections in rodents in former Czechoslovakia. *Bartonella* sp. belonging to lineage 3 was identified by molecular methods in the spleen of an *A. flavicollis* female from central Slovakia in 2006 (Špitalská et al. 2008). Furthermore, Karbowiak et al. (2009) detected *Bartonella* in erythrocytes in Giemsa-stained blood smears of *A. agrarius* from two sites in eastern Slovakia. Recently, Kraljik et al. (2016) ascribed *Bartonella* genotypes in *A. agrarius* from eastern Slovakia to *B. grahamii*, *B. taylorii*, *B. birtlesii*, *B. clarridgeiae/B. rochalimae* and *B. elizabethae/B. tribocorum*. Despite these studies, *Bartonella* epidemiology and host-vector associations in Central Europe, including Slovakia, are not well understood. 4 5

The present study analyses the prevalence and diversity of *Bartonella* spp. in small mammals from three ecologically distinct habitats of south-western and central Slovakia. The presence of *Bartonella* was examined in rodents as well as in questing *Ixodes ricinus* (L., 1758) to determine the role of ticks in transmission. 6

Materials and methods 7

Small mammals and ticks 8

Small mammals were live-trapped at three sites in Slovakia. The two sites, Bratislava (202–334 m above sea level, asl), which represents a suburban forest park with significant anthropogenic impact, and Fúgelka (336–436 m asl), which comprises a non-fragmented deciduous forest, are located in the Small Carpathian Mountains (south-western Slovakia). The mountains have densely forested regions that are dominated by sessile oak (*Quercus petraea*) and European hornbeam (*Carpinus betulus*) at lower altitudes and European beech (*Fagus sylvatica*) at higher altitudes. The Bratislava and Fúgelka study sites were described in detail by Kazimírová et al. (2016). The third site, Prievidza (central Slovakia; 289 m asl), is near a settlement and represents a forest-steppe rural area with Carpathian oak-hornbeam forests (Berthová et al. 2016). 9

A total of 640 rodents comprising six species (*A. flavicollis*, wood mouse *Apodemus sylvaticus* (L., 1758), *My. glareolus*, common vole *Microtus arvalis* (Pallas, 1778), European pine vole *Microtus subterraneus* (de Selys-Longchamps, 1836), and Eurasian harvest mouse *Micromys minutus* (Pallas, 1771)) were live-trapped from April–July and September–October 2012–2014 by Sherman live traps baited with oat flakes (for details, see Svitálková et al. 2015 and Minichová et al. 2017). Captured rodents were transported to the laboratory, anaesthetized and sacrificed humanely according to the current laws of the Slovak Republic. A fresh intact corpse of a European mole *Talpa europaea* L., 1758 was found in the suburban forest park in June 2012. Spleen tissue samples were obtained from each animal and stored in 70% ethanol for DNA extraction. 10

In addition, 5767 questing *I. ricinus*, including 4795 nymphs and 972 adults, were collected by dragging the vegetation in Bratislava and Fúgelka. The ticks were stored in 11

70% ethanol and subsequently identified by species, developmental stage and sex (see Svitálková et al. 2015). A selection of 1335 questing ticks (194 nymphs, 191 females and 90 males from Bratislava; 292 nymphs, 278 females and 290 males from Fúgelka) in pools of 3–7 samples were screened for the presence of *Bartonella.*

Molecular analysis

DNA samples from rodent spleens and questing ticks were used from previous studies (Svitálková et al. 2015; Minichová et al. 2017). We used PCR to determine if *Bartonella* DNA was present in individual rodent spleen samples and pooled tick samples. The 16S–23S rRNA intergenic spacer region (ITS) was amplified, generating a 420–780 bp product (Maggi et al. 2006). PCR reactions were carried out in a total volume of 20 μl containing 2 μl of DNA template, 10 μl of Master mix (SuperHot PCR Master Mix, Bioron, Germany), 0.2 μl of each primer (10 μM), 0.5 μl of 100 mM $MgCl_2$ and 7.1 μl of nuclease free water. Amplification was performed in a thermocycler (Labcycler, SensoQuest, Germany). The thermal cycle reaction consisted of an initial 2 min denaturation at 95 °C, followed by 40 cycles at 94 °C for 30 s, 66 °C for 30 s and 72 °C for 30 s, with a final extension at 72 °C for 5 min. Positive and negative controls consisted of *Bartonella* DNA from infected rodents confirmed by sequencing and sterile water, respectively, and were used in each PCR reaction. PCR products were visualised on 1% agarose gels stained with GelRed Nucleic Acid Gel Stain (Biotium, Hayward, Canada). *Bartonella*-positive amplicons were purified and then analysed by sequencing in both directions with the same primers as for PCR amplification by Macrogen Inc. (Amsterdam, The Netherlands). DNA sequences were compared with those available in GenBank using the Basic Local Alignment Search Tool (BLAST) (http://blast.ncbi.nlm.nih.gov). The new sequences were submitted to GenBank under accession numbers MF039545-MF039578. A phylogenetic tree was constructed using the neighbour-joining method (Saitou and Nie 1987). Phylogenetic analysis was conducted using MEGA5 software (Tamura et al. 2011).

Statistical analysis

Statistical analyses to test for differences in the prevalence of bartonellae in rodents between years, seasons and sites were carried out using Fisher's exact test with an online calculator (http://www.socscistatistics.com). A p value < 0.05 was considered significant. Ninety-five percent confidence intervals (CI) were calculated using an online calculator (http://epitools.ausvet.com.au).

Results

Prevalence of *Bartonella* species

DNA of *Bartonella* spp. was not confirmed in any of the questing *I. ricinus* samples. In contrast, out of 641 small mammals represented by seven species, *Bartonella* DNA was detected in the spleens of 415 animals, including the European mole. The overall infection rate in rodents was 64.8% (95% CI 61.1–68.5), and ranged from 56.0% (95% CI 50.2–61.7) in the suburban habitat to 73.8% (95% CI 68.7–78.8) in the non-fragmented forest (Table 1). *Bartonella* spp. were detected in 63.0% (95% CI 58.2–67.9) of *A. flavicollis*, 69.0% (95% CI 63.0–74.9) of *My. glareolus* and 61.1% (95% CI 38.6–79.7) of *M. arvalis.* We found a significant difference in the bacterial prevalence between the Bratislava and Fúgelka sites ($p < 0.05$). Due to the small number of samples from 2013, we only investigated differences in prevalence between 2012 and 2014. The difference in *Bartonella* prevalence between 2012 and 2014 was significant in Bratislava ($p = 0.0033$) but not in Fúgelka ($p = 0.5456$). We detected an increase in *Bartonella*-positive rodents in autumn (86.1%; 95% CI 82.3–89.8) compared to spring (41.6%; 95% CI 36.0–47.1) (Table 2), and this difference between seasons was significant ($p < 0.05$).

Diversity of *Bartonella* in small mammals

Sequence analysis of the 16S–23S rRNA intergenic spacer region in 153 randomly selected spleen DNA lysates showed that the *Bartonella* strains detected among the investigated rodents, and the European mole belonged to multiple clusters, including *B. taylorii*, *B. grahamii*, *B. elizabethae*, *Bartonella* sp. wbs11 and *B. rochalimae. Bartonella* spp. in rodents from Fúgelka and Prievidza belonged to four different clusters and thus were more diverse than in the small mammals from Bratislava, where only two clusters (*B. taylorii* and *B. rochalimae*) were detected. The majority of DNA sequences were ascribable to *B. taylorii* (Fig. 1). Sequences from this study and their GenBank Accession numbers are given in Table S1.

Forty-eight sequences derived from spleen samples of small mammals trapped in Bratislava (*A. flavicollis*: 12 males, 13 females; *My. glareolus*: 10 males, 12 females; 1 *T. europaea*), 70 sequences from Fúgelka (*A. flavicollis*: 22 males, 13 females; *My. glareolus*: 16 males, 14 females; *M. arvalis*: 4 females, 1 male) and 12 sequences from Prievidza (*A. flavicollis*: 3 males, 5 females; *My. glareolus*: 1 male, 3 females) were identified as *B. taylorii.* Identities of these sequences ranged from 90 to 100% of bacterial sequences (GenBank Acc. Nos AJ269788, AJ269784) were previously identified in rodents from the United Kingdom (Birtles et al. 2000).

Table 1 Prevalence of *Bartonella* spp. in rodents caught at three sites in Slovakia in 2012–2014 [no. of infected/no. of captured (prevalence %)]

Site/habitat type Year	Rodent species					Total
	Apodemus flavicollis	*Apodemus sylvaticus*	*Myodes glareolus*	*Microtus arvalis*	*Micromys minutus*	
Bratislava/suburban						
2012	36/87 (41.4)	0/1	54/94 (57.5)			90/182 (49.4)
2013	1/3 (33.3)					1/3 (33.3)
2014	55/84 (65.5)		18/24 (75.0)			73/108 (67.6)
Subtotal	92/174 (52.9)	0/1	72/118 (61.0)			164/293 (56.0)
Fúgelka/sylvatic						
2012	77/116 (66.4)	0/1	72/79 (91.1)	11/18 (61.1)		160/214 (74.8)
2014	42/52 (81.8)		12/23 (52.0)		0/1	54/76 (71.0)
Subtotal	119/168 (70.8)	0/1	84/102 (82.4)	11/18 (61.1)	0/1	214/290 (73.8)
Prievidza/rural						
2012	31/40 (77.5)		4/12 (33.3)			35/52 (67.3)
2013	2/5 (40.0)					2/5 (40.0)
Subtotal	33/45 (73.3)		4/12 (33.3)			37/57 (64.9)
Total	244/387 (63.0)	0/2	160/232 (69.0)	11/18 (61.1)	0/1	415/640 (64.8)

Two sequences, one from the spleen of an *A. flavicollis* male in Fúgelka and another from an *A. flavicollis* female from Prievidza were 95–99% identical to uncultured *Bartonella* sp. wbs11 identified in the blood of *A. sylvaticus* from the United Kingdom (AJ269792) (Birtles et al. 2000). *Bartonella* sp. wbs11 is 90% identical to *B. doshiae* (AJ269786) and is a rodent-associated species.

In the third cluster, four sequences (derived from a male and female *A. flavicollis* and a male and female *My. glareolus* from Fúgelka) were 99% identical to *B. grahamii* (AJ269785), which was previously identified in the blood of woodland rodents in the United Kingdom (Birtles et al. 2000).

A sequence derived from an *A. flavicollis* female from Prievidza was 96% identical to a *B. elizabethae* (JF766264) sequence that was isolated from *Meriones libycus* heart tissue in Georgia (Malania et al. 2016).

Seven sequences derived from rodents in Fúgelka (*A. flavicollis*: 4 females, 1 male; *M. arvalis*: 1 male, 1 female), two from *A. flavicollis* females from Bratislava, and six from rodents from Prievidza (*A. flavicollis*: 2 females, 4 males) were 98–100% identical to an uncultured *Bartonella* sp. (DQ155376) that was found in *A. agrarius* from Slovenia (Knap et al. 2007). Two sequences from rodents trapped in Fúgelka (MF039562 and MF039562, see Table S1) were 87 and 90% identical, respectively, to uncultured *Bartonella* sp. (DQ155376). The ITS region sequences of this uncultured *Bartonella* sp. (DQ155376) were similar to those of *B. rochalimae* (87% identity; KU292577) and *B. clarridgeiae* (91% identity; AF312497).

Table 2 Seasonal prevalence of *Bartonella* spp. in rodents caught at three sites in Slovakia in 2012–2014 [no. of infected/no. of captured (prevalence %)]

Season	Month	Site/habitat type			Total
		Bratislava/suburban	Fúgelka/sylvatic	Prievidza/rural	
Spring	April		10/25 (40.0)		10/25 (40.0)
	May	0/1		2/3 (66.7)	2/4 (50.0)
	June	72/193 (37.3)	42/81 (501.9)		114/274 (41.6)
	Subtotal	72/194 (37.1)	52/106 (49.1)	2/3 (66.7)	126/303 (41.6)
	July			0/1	0/1
Autumn	September	93/100 (93.0)	70/79 (88.6)	35/53 (66.0)	198/232 (85.3)
	October		92/105 (87.6)		92/105 (87.6)
	Subtotal	93/100 (93.0)	162/184 (88.0)	35/53 (66.0)	290/337 (86.1)

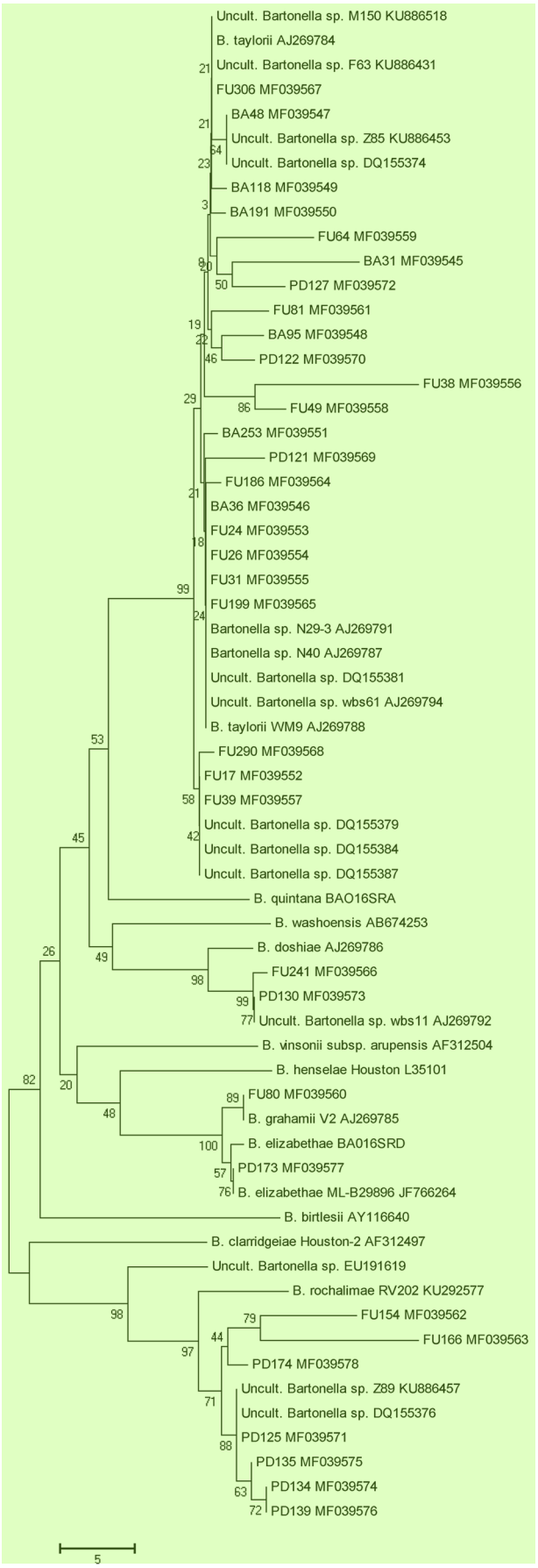


Fig. 1 Phylogenetic tree inferred from comparison of the *Bartonella* 16S–23S internal transcribed spacer sequences. The evolutionary history was inferred using the neighbour-joining method. The percentages of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) are shown next to the branches (Felsenstein, 1985). GenBank accession numbers are included. For more information on the source (host, locality) of the analysed sequences derived from the present study, see Table S1

Discussion

Ectoparasitic arthropods and small mammals play important roles in the circulation of *Bartonella* species in Europe (Regier et al. 2016). Here, we detected the presence of *Bartonella* in small mammals from three habitat types in south-western and central Slovakia, but failed to detect bacteria in questing *I. ricinus* ticks.

The role of ticks in the epidemiology of *Bartonella* is unclear (Harrison et al. 2012; Regier et al. 2016; Silaghi et al. 2016), although experimental vector competence studies have demonstrated the transmission of *B. henselae* and *B. birtlesii* by *I. ricinus* ticks (Cotté et al. 2008; Reis et al. 2011). Our findings are supported by studies conducted in Sweden, Great Britain, Hungary, central Slovakia, Germany, Finland and Serbia, where *Bartonella* was not detected in questing ticks (Billeter et al. 2008; Špitalská et al. 2008; Schorn et al. 2011; Sormunen et al. 2016; Potkonjak et al. 2016). In contrast, up to 60% prevalence of *Bartonella* was detected in questing *I. ricinus* from particular sites in Europe (Regier et al. 2016). Moreover, *Bartonella* has been detected in engorged ticks from different hosts. For example, Lipatova et al. (2015) identified *B. grahamii* in *I. ricinus* larvae that were feeding on small mammals in Lithuania. Recently, Regier et al. (2017) showed that *I. ricinus* ticks that were removed from cats were positive for *B. henselae* DNA. These results indicate that ticks become infected by feeding on infected hosts and that pathogens can survive within ticks, but do not prove that ticks can transmit the infection to a mammalian host. Thus, further research is necessary to elucidate the role of ticks in the transmission of *Bartonella* spp. Rudolf et al. (2016) also highlighted the necessity to extend the list of haematophagous arthropods that are involved in *Bartonella* transmission, as they have demonstrated infection of sheep keds (Diptera) with *B. melophagi*.

Relatively high infection rates (from 25% to over 80%) and a great diversity of *Bartonella* spp. have been reported in wild-living small rodent populations worldwide (Gutiérrez et al. 2015). In our study, *Bartonella* infection was demonstrated in 64.8% of examined rodents. This percentage is similar to levels reported from the United Kingdom (62%; Birtles et al. 1994) and Germany (65.8%; Silaghi et al. 2016) but is higher than the rates found in other European sites, e.g. 25% in Great Britain (Baker 1974; Healing 1981), 25–31% in Poland (Bajer

et al. 2001; Pawełczyk et al. 2004; Welc-Falęciak et al. 2008), 27–28% in Denmark (Engbaek and Lawson 2004) and Spain (Maggi et al. 2009; Gil et al. 2010), 30% in Greece (Tea et al. 2004), 40.4% in Slovenia (Knap et al. 2007) and 30–56.4% in France (Gundi et al. 2004; Buffet et al. 2012). In contrast, a *Bartonella* prevalence rate of over 80% was detected in rodents in southern China and northern USA (Ying et al. 2002; Bai et al. 2011). 1

In addition to spatial variations, interspecific differences and temporal changes in *Bartonella* prevalence have been observed in rodents (Wiger 1979; Bajer et al. 2001; Karbowiak et al. 2005). In our study, the highest total prevalence was detected in *My. glareolus* (69.0%) and the lowest was in *M. arvalis* (61%), although the values differed between sites and years (see Table 1). The range of interspecific variation in *Bartonella* spp. prevalence may be even greater, as demonstrated in rodents from Poland (7–66%) (Karbowiak et al. 2005) and Norway (10–32%) (Wiger 1979). These findings suggest that prevalence data may be affected by the density and dynamics of the rodent populations as well as by the numbers of examined animals belonging to a particular species. Furthermore, seasonal changes in *Bartonella* prevalence have also been registered. For example, Pawełczyk et al. (2004) found that Bartonella infections in the blood of *M. arvalis* in Poland were rare in the spring but were common in summer and autumn. Similar results were obtained in our study, i.e. a higher prevalence of infection was recorded in autumn than in spring. This seasonal difference was not detected at the rural site in Prievidza ($p = 1.0$), which may be due to the low sample size. 2

Rodent-associated *Bartonella* species are genetically diverse (Gutiérrez et al. 2015; Regier et al. 2016). The *Bartonella* spp. present in the rodents and European mole investigated in this study were found to belong to multiple groups, including *B. taylorii*, *B. grahamii*, *B. elizabethae*, *Bartonella* sp. wbs11 and *B. rochalimae*. These species were clustered into phylogenetic lineages 3 and 4 and were also confirmed in *A. agrarius* from eastern Slovakia (Kraljik et al. 2016). Likewise, Buffet et al. (2012) detected different *Bartonella* genotypes that were related to three known species (*B. taylorii*, *B. grahamii* and *B. doshiae*) and two unidentified *Bartonella* species that were associated with rodents in France. *B. taylorii*, *B. grahamii* and *B. rochalimae* were also found in rodents in Lithuania (Lipatova et al. 2015) and uncultured *Bartonella* sp., *B. grahamii*, *B. taylorii* and *Bartonella* sp. N40 were detected in rodents from Germany (Silaghi et al. 2016). 3

In this study, the majority of sequences derived from rodent spleens (85%) were ascribable to *B. taylorii*. This species causes infection in animals, although its real pathogenic potential is unknown (Lipatova et al. 2015). We expected to find *B. taylorii* and detected it in *A. flavicollis*, *My. glareolus*, *M. arvalis* and *T. europaea*. Previously, *B. taylorii* was isolated from *My. glareolus* in the United Kingdom (Bown et al. 2004) and Siberia (Jiyipong et al. 2014) and was detected in *A. sylvaticus*, *A. agrarius*, *A. uralensis* (Pallas, 1811), *A. witherbyi* (Thomas, 1902), *M. minutus*, *Microtus agrestis* (L., 1761), *Microtus* spp. and *Mus domesticus* L., 1758 in Austria (Schmidt et al. 2014), Lithuania (Lipatova et al. 2015), Sweden (Holmberg et al. 2003), Spain (Gil et al. 2010) and Greece (Tea et al. 2004). 4 5

Four sequences derived from *A. flavicollis* and *My. glareolus* in the Fúgelka natural habitat were found to be similar to *B. grahamii*, a causative agent of human renitis (Kerkhoff et al. 1999). *B. grahamii* was also isolated from a human patient in the USA in a region that is endemic to this pathogen (Eremeeva et al. 2007) and was found to infect different rodent species (*A. sylvaticus*, *A. agrarius*, *A. uralensis*, *A. witherbyi*, *M. minutus*, *M. arvalis*, *M. agrestis* and *M. domesticus*) (Mogollon-Pasapera et al. 2009; Oksi et al. 2013; Lipatova et al. 2015). Because of its worldwide occurrence and its prevalence in Slovakia, *B. grahamii* may represent a potential risk for human infection in this country. 6

We also detected *B. elizabethae* DNA in *A. flavicollis*. This result is surprising because this species is a human pathogen and has been associated with rats. It was described in 1993 in the USA, causes endocarditis and is transmitted by fleas (Mogollon-Pasapera et al. 2009). However, Kraljik et al. (2016) detected *B. elizabethae* in wild-living mice *A. agrarius* from eastern Slovakia. *B. elizabethae* was also cultivated from *A. sylvaticus* in Georgia (Malania et al. 2016). Although it is unlikely to represent a serious risk for the local human population, it should be considered a potential agent of infection to people in contact with rodents. 7

Fifteen sequences found primarily in *A. flavicollis* but also in *M. arvalis* were clustered into phylogenetic lineage 3 and showed identity to *B. rochalimae*, an infectious zoonotic agent. This species is associated with dogs and foxes (Engel et al. 2011), although it has been suggested that rodents are reservoirs for this pathogen (Lin et al. 2008). Our data support previous identifications of *B. rochalimae* in *My. glareolus* in Lithuania (Lipatova et al. 2015) and France (Buffet et al. 2012). 8

In conclusion, our study provides evidence that a high proportion of rodents from selected sites in south-western and central Slovakia harbour *Bartonella*, which can be transmitted to other hosts including humans. Thus, it is important to continue investigating *Bartonella* spp. that circulate in the wildlife to estimate potential risks for the Slovakian population that have contact with rodents. Small mammals and humans share the same environment and thus extensive surveillance should be performed, especially in areas where humans encounter small mammals. 9

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

All procedures performed in studies involving animals were in accordance with the ethical standards of the institution or practice at which the studies were conducted.

Work with small mammals was approved by the Ministry of Environment of the Slovak Republic, Regional Environmental Office in Bratislava (licence ZPO-594/2012-SAB).

This article does not contain any studies with human participants performed by any of the authors.

References

Bai Y, Calisher CH, Kosoy MY, Root JJ, Doty JB (2011) Persistent infection or successive reinfection of deer mice with *Bartonella vinsonii* subsp. *arupensis*. Appl Environ Microbiol 77:1728–1731

Bajer A, Pawełczyk A, Behnke J, Gilbert F, Siński E (2001) Factors affecting the component community structure of haemoparasites in bank voles (*Clethrionomys glareolus*) from the Mazury Lake District region of Poland. Parasitology 122:43–54

Baker JR (1974) Protozoan parasites of the blood of British wild birds and mammals. J Zool London 172:169–190

Berthová L, Slobodník V, Slobodník R, Olekšák M, Sekeyová Z, Svitálková Z, Kazimírová M, Špitalská E (2016) The natural infection of birds and ticks feeding on birds with *Rickettsia* spp. and *Coxiella burnetii* in Slovakia. Exp Appl Acarol 68:299–314

Billeter SA, Levy MG, Chomel BB, Breitschwerdt EB (2008) Vector transmission of *Bartonella* species with emphasis on potential for tick transmission. Med Vet Entomol 22:1–15

Billeter SA, Sangmaneedet S, Kosakewich RC, Kosoy MY (2012) *Bartonella* species in dogs and their ectoparasites from Khon Kaen Province, Thailand. Southeast Asian J Trop Med Public Health 43:1186–1192

Birtles RJ, Harrison TG, Molyneux DH (1994) *Grahamella* in small woodland mammals in the U.K.: isolation, prevalence and host specificity. Ann Trop Med Parasitol 88:317–327

Birtles RJ, Harrison TG, Saunders NA MDH (1995) Proposals to unify the genera *Grahamella* and *Bartonella*, with descriptions of *Bartonella taplae* comb. nov., *Bartonella peromysci* comb. nov., and three new species, *Bartonella grahamii* sp. nov., *Bartonella taylorii* sp. nov., and *Bartonella doshiae* sp. nov. Int J Syst Bacteriol 45:1–8

Birtles RJ, Hazel S, Bown K, Raoult D, Begon M, Bennett M (2000) Subtyping of uncultured bartonellae using sequence comparison of 16S/ 23S rRNA intergenic spacer regions amplified directly from infected blood. Mol Cell Probes 14:79–87

Blaňarová L, Stanko M, Miklisová D, Víchová B, Mošanský L, Kraljik J, Bona M, Derdáková M (2016) Presence of *Candidatus* Neoehrlichia mikurensis and *Babesia microti* in rodents and two tick species (*Ixodes ricinus* and *Ixodes trianguliceps*) in Slovakia. Ticks Tick Borne Dis 7:319–326. https://doi.org/10.1016/j.ttbdis.2015.11.008

Bown KJ, Bennett M, Begon M (2004) Flea-borne *Bartonella grahamii* and *Bartonella taylorii* in bank voles. Emerg Infect Dis 10(4):684e7

Breitschwerdt EB, Kordick DL (2000) *Bartonella* infection in animals: carriership, reservoir potential, pathogenicity, and zoonotic potential for human infection. Clin Microbiol Rev 13:428–438

Buffet JP, Marsot M, Vaumourin E, Gasqui P, Masséglia S, Marcheteau E, Huet D, Chapuis JL, Pisanu B, Ferquel E, Halos L, Vourc'h G, Vayssier-Taussat M (2012) Co-infection of *Borrelia afzelii* and *Bartonella* spp. in bank voles from a suburban forest. Comp Immunol Microbiol Infect Dis 35:583–589

Cotté V, Bonnet S, Le Rhun D, Le Naour E, Chauvin A, Boulouis HJ, Lecuelle B, Lilin T, Vayssier-Taussat M (2008) Transmission of *Bartonella henselae* by *Ixodes ricinus*. Emerg Infect Dis 14:1074–1080

Daly JS, Worthington MG, Brenner DJ, Moss CW, Hollis DG, Weyant RS, Steigerwalt AG, Weaver RE, Daneshvar MI, O'Connor SP (1993) *Rochalimaea elizabethae* sp. nov. isolated from a patient with endocarditis. J Clin Microbiol 31:872–881

Engbaek K, Lawson PA (2004) Identification of *Bartonella* species in rodents, shrews and cats in Denmark: detection of two *Bartonella henselae* variants, one in cats and the other in the long-tailed field mouse. APMIS 112:336–341

Engel P, Salzburger W, Liesch M, Chang CC, Maruyama S, Lanz C, Calteau A, Lajus A, Médigue C, Schuster SC, Dehio C (2011) Parallel evolution of a type IV secretion system in radiating lineages of the host-restricted bacterial pathogen *Bartonella*. PLoS Genet 7:e1001296

Eremeeva ME, Gerns HL, Lydy SL et al (2007) Bacteremia, fever, and splenomegaly caused by a newly recognized *Bartonella* species. N Engl J Med 356:2381e7

Felsenstein J (1985) Confidence limits on phylogenies: an approach using the bootstrap. Evolution 39:783–791

Gil H, García-Esteban C, Barandika JF, Peig J, Toledo A, Escudero R, Jado I, Rodríguez-Vargas M, García-Amil C, Lobo B, Roales P, Rodríguez-Moreno I, Olmeda AS, García-Pérez AL, Anda P (2010) Variability of *Bartonella* genotypes among small mammals in Spain. Appl Environ Microbiol 76(24):8062–8070

Gundi VA, Davoust B, Khamis A, Boni M, Raoult D, La Scola B (2004) Isolation of *Bartonella rattimassiliensis* sp. nov. and *Bartonella phoceensis* sp. nov. from European *Rattus norvegicus*. J Clin Microbiol 42:3816–3818

Gutiérrez R, Krasnov B, Morick D, Gottlieb Y, Khokhlova IS, Harrus S (2015) *Bartonella* infection in rodents and their flea ectoparasites: an overview. Vector Borne Zoonotic Dis 15(1):27–39

Hamšíková Svitálková Z, Haruštiaková D, Mahríková L, Mojšová M, Berthová L, Slovák M, Kocianová E, Vayssier-Taussat M, Kazimírová M (2016) *Candidatus* Neoehrlichia mikurensis in ticks and rodents from urban and natural habitats of South-Western Slovakia. Parasit Vectors 9:2

Hamšíková Z, Kazimírová M, Haruštiaková D, Mahríková L, Slovák M, Berthová L, Kocianová E, Schnittger L (2016) *Babesia* spp. in ticks and wildlife in different habitat types of Slovakia. Parasit Vectors 9:292

Hamšíková Z, Coipan C, Mahríková L, Minichová L, Sprong H, Kazimírová M (2017) *Borrelia miyamotoi* and co-infection with *Borrelia afzelii* in *Ixodes ricinus* ticks and rodents from Slovakia. Microb Ecol 73:1000–1008

Hanincová K, Schäfer SM, Etti S, Sewell HS, Taragelová V, Ziak D, Labuda M, Kurtenbach K (2003) Association of *Borrelia afzelii* with rodents in Europe. Parasitology 126:11–20

Harrison A, Bown KJ, Montgomery WI, Birtles RJ (2012) *Ixodes ricinus* is not an epidemiologically relevant vector of *Bartonella* species in

the wood mouse (*Apodemus sylvaticus*). Vector Borne Zoonotic Dis 12(5):366–371

Healing TD (1981) Infections with blood parasites in the small British rodents *Apodemus sylvaticus*, *Clethrionomys glareolus* and *Microtus agrestis*. Parasitology 83:179–189

Holmberg M, Mills JN, McGill S, Benjamin G, Ellis (2003) Bartonella infection in sylvatic small mammals of central Sweden. Epidemiol Infect 130:149–157

Jiyipong T, Jittapalapong S, Morand S, Rolain JM (2014) *Bartonella* species in small mammals and their potential vectors in Asia. Asian Pac J Trop Biomed 4(10):757e67

Karbowiak G, Rychlik L, Nowakowski W, Wita I (2005) Natural infections of small mammals with blood parasites on the borderland of boreal and temperate forest zones. Acta Theriol 50:31–42

Karbowiak G, Stanko M, Fričová J, Wita I, Hapunik J, Peťko B (2009) Blood parasites of the striped field mouse *Apodemus agrarius* and their morphological characteristics. Biologia 64:1219–1224

Kazimírová M, Hamšíková Z, Kocianová E, Marini G, Mojšová M, Mahríková L, Berthová L, Slovák M, Rosá R (2016) Relative density of host-seeking ticks in different habitat types of south-western Slovakia. Exp Appl Acarol 69:205–224

Kerkhoff FT, Bergmans AM, van Der Zee A, Rothova A (1999) Demonstration of *Bartonella grahamii* DNA in ocular fluids of a patient with neuroretinitis. J Clin Microbiol 37:4034–4038

Knap N, Duh D, Birtles R, Trilar T, Petrovec M, Avšič-Županc T (2007) Molecular detection of *Bartonella* species infecting rodents in Slovenia. FEMS Immunol Med Microbiol 50(1):45–50

Kosoy M, Murray M, Gilmore RD Jr, Bai Y, Gage KL (2003) *Bartonella* strains from ground squirrels are identical to *Bartonella washoensis* isolated from a human patient. J Clin Microbiol 41:645–650

Kosoy M, Bai Y, Sheff K, Morway C, Baggett H, Maloney SA, Boonmar S, Bhengsri S, Dowell SF, Sitdhirasdr A, Lerdthusnee K, Richardson J, Peruski LF (2010) Identification of *Bartonella* infections in febrile human patients from Thailand and their potential animal reservoirs. Am J Trop Med Hyg 82:1140–1145

Kraljik J, Paziewska-Harris A, Miklisová D, Blaňarová L, Mošanský L, Bona M, Stanko M (2016) Genetic diversity of *Bartonella* genotypes found in the striped field mouse (*Apodemus agrarius*) in Central Europe. Parasitology 143(11):1437–1442

Lin JW, Chen CY, Chen WC, Chomel BB, Chang CC (2008) Isolation of *Bartonella* species from rodents in Taiwan including a strain closely related to '*Bartonella rochalimae*' from *Rattus norvegicus*. J Med Microbiol 57:1496e501

Lipatova I, Paulauskas A, Puraite I, Radzijevskaja J, Balciauskas L, Gedminas V (2015) *Bartonella* infection in small mammals and their ectoparasites in Lithuania. Microbes Infect 17:884–888

Maggi RG, Diniz PP, Cadenas MB, Breitschwerdt EB (2006) The use of molecular diagnostic techniques to detect *Anaplasma*, *Bartonella* and *Ehrlichia* species in arthropods or patients. The International Canine Vector-Borne Disease Symposium; April 18-20th 2006 Billesley, Alcester, UK: 9–14

Maggi RG, Kosoy M, Mintzer M, Breitschwerdt EB (2009) Isolation of Candidatus *Bartonella melophagi* from human blood. Emerg Infect Dis 15:66–68

Malania L, Bai Y, Osikowicz LM, Tsertsvadze N, Katsitadze G, Imnadze P, Kosoy M (2016) Prevalence and diversity of *Bartonella* species in rodents from Georgia (Caucasus). Am J Trop Med Hyg 95:466–471

Minichová L, Hamšíková Z, Mahríková L, Slovák M, Kocianová E, Kazimírová M, Škultéty Ľ, Štefanidesová K, Špitalská E (2017) Molecular evidence of *Rickettsia* spp. in ixodid ticks and rodents in suburban, natural and rural habitats in Slovakia. Parasit Vectors 10:158. https://doi.org/10.1186/s13071-017-2094-8

Mogollon-Pasapera E, Otvos L, Giordano A, Cassone M (2009) *Bartonella*: Emerging pathogen or emerging awareness? Int J Infect Dis 13:3–8

Morozova OV, Chernousova N, Morozov IV (2005) Detection of the *Bartonella* DNA by the method of nested PCR in patients after tick bites in Novosibirsk region. Mol Gen Mikrobiol Virusol 4:14–17

Mosepele M, Mazo D, Cohn J (2012) *Bartonella* infection in immunocompromised hosts: immunology of vascular infectionand vasoproliferation. Clin Dev Immunol 2012:612809

Mullins KE, Hang J, Clifford RJ, Onmus-Leone F, Yang Y, Jiang J, Leguia M, Kasper MR, Maguina C, Lesho EP, Jarman RG, Richards A, Blazes D (2017) Whole-genome analysis of *Bartonella ancashensis*, a novel pathogen causing verruga peruana, rural Ancash region, Peru. Emerg Infect Dis 23:430–438

Oksi J, Rantala S, Kilpinen S, Silvennoinen R, Vornanen M, Veikkolainen V et al (2013) Cat scratch disease caused by *Bartonella grahamii* in an immunocompromised patient. J Clin Microbiol 51(8):2781e4

Pawełczyk A, Bajer A, Behnke JM, Gilbert FS, Siński E (2004) Factors affecting the component community structure of haemoparasites in common voles (*Microtus arvalis*) from the Mazury Lake District region of Poland. Parasitol Res 92:270–284

Potkonjak A, Gutiérrez R, Savić S, Vračar V, Nachum-Biala Y, Jurišić A, Kleinerman G, Rojas A, Petrović A, Baneth G, Harrus S (2016) Molecular detection of emerging tick-borne pathogens in Vojvodina, Serbia. Ticks Tick Borne Dis 7:199–203

Regier Y, O'Rourke F, Kempf VAJ (2016) *Bartonella* spp.—a chance to establish One Health concepts in veterinary and human medicine. Parasit Vectors 9:261

Regier Y, Ballhorn W, Kempf VAJ (2017) Molecular detection of *Bartonella henselae* in 11 *Ixodes ricinus* ticks extracted from a single cat. Parasit Vectors 10:105

Reis C, Cote M, Le Rhun D, Lecuelle B, Levin ML, Vayssier-Taussat M, Bonnet SI (2011) Vector competence of the tick *Ixodes ricinus* for transmission of *Bartonella birtlesii*. PLoS Negl Trop Dis 5:e1186

Rizzoli A, Silaghi C, Obiegala A, Rudolf I, Hubálek Z, Földvári G, Plantard O, Vayssier-Taussat M, Bonnet S, Špitalská E, Kazimírová M (2014) *Ixodes ricinus* and its transmitted pathogens in urban and peri-urban areas in Europe: new hazards and relevance for public health. Front Public Health 2:251

Rudolf I, Betášová L, Bischof V, Venclíková K, Blažejová H, Mendel J, Hubálek Z, Kosoy M (2016) Molecular survey of arthropod-borne pathogens in sheep keds (*Melophagus ovinus*), Central Europe. Parasitol Res 115:3679

Saitou N, Nie M (1987) The neighbour-joining method: a new method for reconstructing phylogenetic trees. Mol Biol Evol 4:406–425

Schmidt S, Essbauer SS, Mayer-Scholl A, Poppert S, Schmidt-Chanasit J, Klempa B, Henning K, Schares G, Groschup MH, Spitzenberger F, Richter D, Heckel G, Ulrich RG (2014) Multiple infections of rodents with zoonotic pathogens in Austria. Vector Borne Zoonot Dis 14(7):467–475

Schorn S, Pfister K, Reulen H, Mahling M, Silaghi C (2011) Occurrence of *Babesia* spp., *Rickettsia* spp. and *Bartonella* spp. in *Ixodes ricinus* in Bavarian public parks, Germany. Parasit Vectors 4:135

Šebek Z (1960) Některé nové poznatky o krevních parasitech našich hmyzožravců a myšovitých. Vlastivědný Sborník Vysočiny, Oddíl Věd Přírodních, 75–80

Šebek Z (1975) Blutparasiten der wildlebenden Kleinsäuger in der Tschechoslowakei. Folia Parasitol (Praha) 22:11–20

Silaghi C, Pfeffer M, Kiefer D, Kiefer M, Obiegala A (2016) *Bartonella*, rodents, fleas and ticks: a molecular field study on host-vector-pathogen associations in Saxony, Eastern Germany. Microb Ecol 72(4):965–974

Sormunen JJ, Penttinen R, Klemola T, Hanninen J, Vuorinen I, Laaksonen M et al (2016) Tick-borne bacterial pathogens in southwestern Finland. Parasit Vectors 9:168

Špitalská E, Boldiš V, Košťanová Z, Kocianová E, Štefanidesová K (2008) Incidence of various tick-borne microorganisms in rodents and ticks of central Slovakia. Acta Virol 52:175–179

Svitálková Z, Haruštiaková D, Mahríková L, Berthová L, Slovák M, Kocianová E, Kazimírová M (2015) *Anaplasma phagocytophilum* prevalence in ticks and rodents in an urban and natural habitat in South-Western Slovakia. Parasit Vectors 8:276

Tamura K, Peterson D, Peterson N, Stecher G, Nie M, Kumar S (2011) MEGA5: molecular evolutionary genetics analysis using maximum likelihood, evolutionary distance, and maximum parsimony methods. Mol Biol Evol 28:2731–2739

Tea A, Alexiou-Daniel S, Papoutsi A, Papa A, Antoniadis A (2004) *Bartonella* species isolated from rodents, Greece. Emerg Infect Dis 10:963–964

Welc-Falęciak R, Paziewska A, Bajer A, Behnke JM, Siński E (2008) *Bartonella* spp. infection in rodents from different habitats in the Mazury Lake District, Northeast Poland. Vector Borne Zoonotic Dis 8:467–474

Welch DF, Carroll KC, Hofmeister EK, Persing DH, Robison DA, Steigerwalt AG, Brenner DJ (1999) Isolation of a new subspecies, *Bartonella vinsonii subsp. arupensis*, from a cattle rancher: identity with isolates found in conjunction with *Borrelia burgdorferi* and *Babesia microti* among naturally infected mice. J Clin Microbiol 37:2598–2601

Wiger R (1979) Seasonal and annual variations in the prevalence of blood parasites in cyclic species of small rodents in Norway with special reference to *Clethrionomys glareolus*. Holarct Ecol 2:169–175

Ying B, Kosoy MY, Maupin GO, Tsuchiya KR, Gage KL (2002) Genetic and ecologic characteristics of *Bartonell*a communities in rodents in southern China. Am J Trop Med Hyg 66:622–627